I.

LIMITS & CONTINUITY

Limits

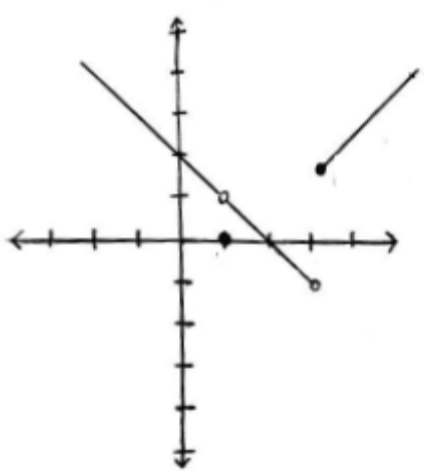
Intuitively, the limit of a function, if it exists, is the value the function approaches as the input approaches some value in the domain.

The mathematical notation for a limit as as follows:

$$\lim_{x \to a} f(x) = L$$

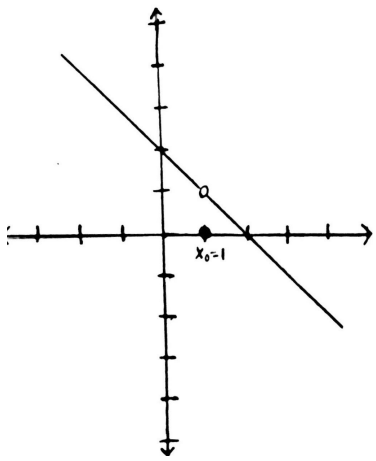
What this means is as x approaches some constant value a, $f(x)$ correspondingly approaches some constant value L.

When can a limit not exist? For instance, let's examine the function below:



What does $f(x)$ approach as x approaches 2? From the left hand side, it appears that $f(x)$ is approaching - 1. However, from the right hand side, $f(x)$ is approaching 2. Therefore, since the limit doesn't approach a single finite value, we say that the limit *does not exist.*

However, if we were asked what $f(x)$ approaches as x approaches, say, -2, we can confidently say that $f(x)$ approaches 4 since the function approaches the same value from both sides. What about this function?



Notice that there exists a *removable discontinuity* at $x = 1$; in other words, $f(x)$ is not *continuous* at $x = 1$. So can we still say that the limit exists?

Actually we can! Remember that the limit is the value a function *approaches* as x approaches some value a. That is, $f(x)$ does not necessarily *have to be* defined at a, it just has to approach some value $f(a)$. Since the function approaches 1 from both sides as x approaches 1, we can say that the limit does indeed exist, and is equal to 1.

However, even though the limit is defined, it isn't actually the equal to the value of $f(1)$. In the graph above, $\lim_{x \to 1} f(x)$ clearly doesn't equal the actual value of $f(1)$, which is 0. This is true for limits in general — the value of the limit does not *necessarily* have to equal the true value at that point.

So, to summarize:

1. A limit $\lim_{x \to a} f(x)$ is said to exist at $x = a$ if $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = L$.

2. If $\lim_{x \to a^+} f(x) \neq \lim_{x \to a^-} f(x)$, then the limit *does not exist* (DNE).

3. $\lim_{x \to a} f(x)$ does not necessarily have to equal $f(a)$.

Computing Limits

If f is continuous on $[a,\ b]$, then it follows that $\lim_{x \to c} f(x) = f(c)$ for any c in the interval $[a,\ b]$.

Essentially, we can compute the limit of any continuous function by simply "plugging in" the value c into $f(x)$.

Sometimes, however, if the function is not continuous everywhere on $[a,\ b]$, the limit $\lim_{x \to c} f(x)$ may not exist.

Such is the case with *jump discontinuities* and *vertical asymptotes*, which we will discuss at a later time. Furthermore, the limit could still exist even if the function is discontinuous — for instance, if a function has *removable discontinuities*. For now, just know that sometimes if the function is not continuous at c, $\lim_{x \to c} f(x)$ may not necessarily exist.

Properties of Limits

Before we go ahead and compute more limits, we need to go over some of the basic properties of limits. We will leave the proofs as an exercise to the reader (Joking — we encourage you to think about it, but the proofs are beyond the scope of what is taught in the AP ® Calculus AB curriculum).

Suppose k is a constant, f and g are functions of x. Assuming that the limit exists, then the following properties hold true:

1. $\lim_{x \to a} k = k$

2. $\lim_{x \to a} k f(x) = k \lim_{x \to a} f(x)$

3. $\lim_{x \to a} f(x) \pm g(x) = \lim_{x \to a} f(x) \pm \lim_{x \to a} g(x)$

4. $\lim_{x \to a} f(x) \cdot g(x) = \lim_{x \to a} f(x) \cdot \lim_{x \to a} g(x)$

5. $\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$, provided that $\lim_{x \to a} g(x) \neq 0$

6. $\lim_{x \to a}[f(x)]^n = [\lim_{x \to a} f(x)]^n$, provided n is an integer.

7. $\lim_{x \to a} b^{f(x)} = b^{\lim_{x \to a} f(x)}$, provided that $b > 0$.

8. $\lim_{x \to a} [\log_b (f(x)] = \log_b [\lim_{x \to a}(f(x)]$, provided that b and $\lim_{x \to a}(f(x)] > 0$.

Limits of Composite Functions

A *composite function* $h(x)$ is one that can be written as the composition of two arbitrary functions $f(g(x))$; that is, the function f is applied to the result of applying the function g to x.

We can think of a composite function $f \circ g(x)$ as g being substituted for x in f.

For example, let $f(x) = 2x^2 + 3x - 2$ and $g(x) = \frac{1}{x}$

We define $f \circ g(x)$ as $2(\frac{1}{x})^2 + 3(\frac{1}{x}) - 2$

Similarly, we define $g \circ f(x)$ as $\frac{1}{2x^2+3x-2}$

If f and g are functions such that $\lim_{x \to c} g(x) = L$ and $\lim_{x \to L} f(x) = f(L)$, then it holds true that $\lim_{x \to c} f(g(x)) = f(\lim_{x \to c} g(x)) = f(L)$.

To summarize:

1. First, find the limit of $g(x)$ (the inside function) approaches as x approaches c.
   a. $\lim_{x \to c} g(x) = L$ (The limit of $g(x)$ is L)

2. Then, find the limit of $f(x)$ (the outside function) approaches as x approaches L.
   a. $\lim_{x \to L} f(x) =$

Limits Algebraically (Indeterminate Forms)

As we discussed earlier, the most simple way of computing a limit is simply just "plugging in" the independent variable into the function. This works some of the time. For instance, $\lim_{x\to 3} x + 1 = 4$ is fairly self-explanatory.

However, sometimes, this yields DNE, where the limit does not exist. Additionally, if this leads to $\frac{0}{0}$ or $\frac{\infty}{\infty}$, the limit is said to be indeterminate, and needs to be worked on further to uncover the true answer (We will discuss in a later chapter).

We could, however, simplify the expressions algebraically to allow us to determine the limit.

<u>Option 1: Factor and Cancel</u>

We could factor the expression and cancel terms, in hopes to remove the part of the expression that yields $\frac{0}{0}$. For example, when we directly substitute in the following problem, it becomes indeterminate.

$$\lim_{x\to 2} \frac{x^2-4}{x-2} = \frac{0}{0}$$

But if we factor and cancel, we get rid of $x-2$, which was causing the problem.

$$\lim_{x\to 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x\to 2} x+2 = 4$$

<u>Option 2: Simplifying Fractions</u>

$$\lim_{x\to 0^-} \frac{\frac{1}{x+4}-\frac{1}{4}}{x} = \lim_{x\to 0^-} \frac{\frac{4}{4(x+4)}-\frac{x+4}{4(x+4)}}{x} = \lim_{x\to 0^-} \frac{\frac{-x}{4(x+4)}}{x} = \lim_{x\to 0^-} \frac{-x}{4(x+4)}\left(\frac{1}{x}\right) = \lim_{x\to 0^-} \frac{-1}{4(x+4)} = \frac{1}{16}$$

Here, we combined the two fractions in the numerator by finding a common denominator. With some simple algebraic manipulation, we can cancel out the $\frac{-x}{x}$, which allows us to plug in 0 and thus determine the final answer. In other problems, make sure to simplify the fractions and use all of the proper fraction rules to find the answer.

Option 3: Rationalization

Finally, we can cancel terms out when there is a square root present. By multiplying by the conjugate, we can get somewhere. Wait, what is the conjugate?

We know that $a^2 - b^2 = (a-b)(a+b)$

Therefore, if we multiply some $(a-b)$ by its conjugate $(a+b)$, we will get $a^2 - b^2$. This can be utilized in some limit problems.

$$\lim_{x\to 8} \frac{\sqrt{x-4}-2}{x-8} \cdot \frac{\sqrt{x-4}+2}{\sqrt{x-4}+2} = \lim_{x\to 8} \frac{x-4-4}{x-8(\sqrt{x-4}+2)} = \lim_{x\to 8} \frac{1}{\sqrt{x-4}+2} = \frac{1}{4}$$

We first multiply by the conjugate (divided by itself to maintain equality). Then, after simplifying, we can cancel out $(x-8)$ which let me finally plug in 8 and get our answer.

Limits at Infinity (End Behavior)

When a limit approaches negative infinity (to the left) or positive infinity (to the right), it is known as end behavior. This tells us how the function behaves as we get further and further towards infinity.

- Does it go to zero?
- Does it shoot upward?
- Does it even exist? Is it unbounded?

Polynomials

A polynomial is a function of the form $f(x) = ax^n + bx^{n-1} + cx^{n-2} + \ldots + dx^2 + ex^1 + f$, where *a, b, c,* etc. are constant non-negative integer *coefficients* and *x* is raised to some integer power *n*. For instance, $f(x) = x^3 + 7x^2 + 12x$ is a polynomial function.

There is an easy way to know a certain end behavior for polynomials. We can tell which way it will go based on the degree and sign of the leading coefficient.

A few quick rules:

- If the highest degree is *even* and the leading coefficient is *positive*, then

  $\lim_{x\to-\infty} f(x) = \lim_{x\to+\infty} f(x) = \infty$

- If the highest degree is *even* and the leading coefficient is *negative*, then

  $\lim_{x\to-\infty} f(x) = \lim_{x\to+\infty} f(x) = -\infty$

- If the highest degree is *odd* and the leading coefficient is *positive*, then

  $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to+\infty} f(x) = \infty$

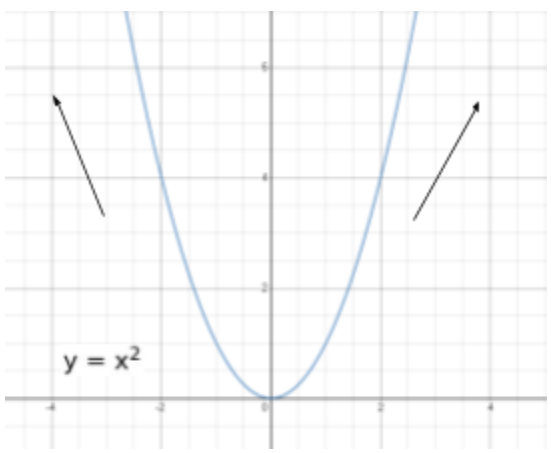
- If the highest degree is *odd* and the leading coefficient is *negative*, then

  $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to+\infty} f(x) = -\infty$

Let's take a look at some example problems:

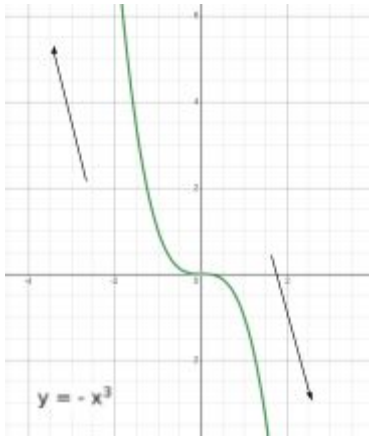
1. $\lim_{x\to\infty} 3x^6 + x^2$

Notice that the leading coefficient is positive and that the highest degree is even. Therefore, $\lim_{x\to\infty} 3x^6 + x^2 = \infty$. We can check by simply plotting the graph, as shown below:



2. $\lim_{x\to\infty} -3x^7 + x^2$

   The leading coefficient is negative, and the highest degree is odd. Therefore, $\lim_{x\to\infty} 3x^6 + x^2 = -\infty$. Again, we can check by simply plotting the graph, as shown below:



<u>Rationals</u>

Suppose that for a continuous function *f*, we knew that $\lim_{x \to \pm\infty} f(x) = \pm\infty$; that is, as x becomes infinitely large, $f(x)$ likewise follows suit.

What do we then know about the following limit $\lim_{x \to \pm\infty} \frac{1}{f(x)}$?

Let's start with something simple. Consider $f(x) = \frac{1}{x}$. What can we conclude about $\lim_{x \to \infty} \frac{1}{x}$?

Let's try plugging in progressively larger values for *x*, and thus try to determine a relationship between *x* and the end behavior of $f(x)$.

$f(1) = \frac{1}{1} = 1$

$f(10) = \frac{1}{10} = .1$

$f(100) = \frac{1}{100} = .01$

$f(1000) = \frac{1}{1000} = .001$

It appears that as *x* gets closer and closer to infinity, then the function $f(x)$ gets closer and closer to 0. Thus, we can say that the limit as x approaches infinity is 0, or $\lim_{x \to \infty} \frac{1}{x} = 0$.

In fact, this is a general rule of thumb for any rational function of the form $f(x) = \frac{c}{x^r}$, where *c* is a constant and $r > 1$.

For any $r > 1$

$\lim_{x \to \infty} \frac{c}{x^r} = 0 \qquad \lim_{x \to -\infty} \frac{c}{x^r} = 0$

Take $f(x) = \frac{1}{x}$

This exercise can be done for negative infinity as well. We encourage you try to prove why $\lim_{x \to -\infty} \frac{1}{x} = 0$.

A *rational function* is any function that can be expressed as $\frac{p(x)}{q(x)}$, where p and q are both polynomials. For instance, $f(x) = \frac{1}{x^2}$ is a polynomial function.

When determining the end behavior for rational functions, we look at the highest degrees of the polynomials in both the numerator and denominator.

There are three scenarios when there's a rational function with degrees on the top and bottom.

1. If $f(x) = \frac{\textit{smaller degree}}{\textit{larger degree}}$, $\lim_{x \to \pm\infty} = 0$
2. If $f(x) = \frac{\textit{same degree}}{\textit{same degree}}$, $\lim_{x \to \pm\infty} = \frac{\textit{leading coefficient}}{\textit{leading coefficient}}$

3a. If $f(x) = \frac{\textit{larger degree}}{\textit{smaller degree}}$, $\lim_{x \to \infty} = \pm\infty$

3b. If $f(x) = \frac{\textit{larger degree}}{\textit{smaller degree}}$, $\lim_{x \to -\infty} = \pm\infty$

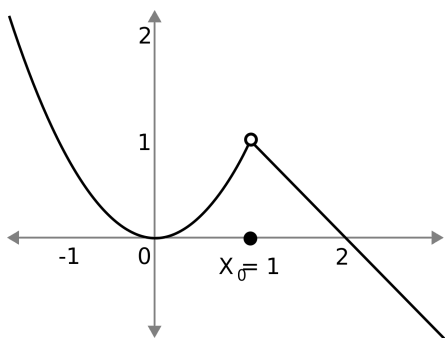
Hint: Plug in large numbers to see how a rational function behaves as it goes left or right

Continuity

A limit only exists at a point $x = a$ *if* $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x)$*: in other words, as* x *approaches* a *from the left and right side,* $f(x)$ *likewise approaches* $f(a)$ *from above and below.*

HOWEVER, for a limit to exist, $\lim_{x \to a} f(x)$ DOES NOT necessarily have to equal $f(a)$. For instance, you could have a removable discontinuity in which *f* is still defined at $x = a$, albeit at a value different from $\lim_{x \to a} f(x)$.

As shown in the diagram below, $\lim_{x \to 1} f(x) \neq f(1)$. The limit at $\lim_{x \to 1} f(x)$ exists, but f is nevertheless discontinuous. It's important to note that $f(a)$ doesn't even have to be defined for the limit to exist.



A function *f* is continuous everywhere if $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = f(a)$ for all points *a* in the domain of *f*. Basically, not only does the limit have to exist for all points *a* on the domain of *f*, the limit must ALSO equal $f(a)$ for all points *a* on the domain of f. *Just think of a continuous function as one you can draw with a pencil without having to lift it up - ever.*

To summarize:

A function *f* is *continuous* at *c* if the following conditions are met:

1. $f(c)$ is *defined*
2. $\lim_{x \to c} f(x)$ *exists*; that is, $\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x)$
3. $\lim_{x \to c} f(x) = f(c)$.

All three of these conditions must be true for f to be continuous at c. This is just one of those things you have to memorize. To put it simply, <u>the function value at c must be defined, the limit at c must exist, and they have to equal each other.</u>

A function f is continuous on the closed interval $[a,\ b]$ if f in continuous on the open interval $(a,\ b)$, and the following holds true for a and b:

- $\lim_{x \to a^-} f(x) = f(a)$
- $\lim_{x \to b^+} f(x) = f(b)$

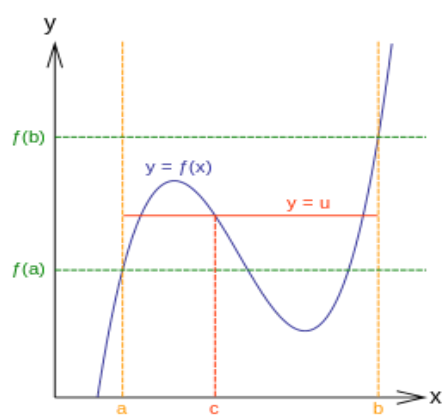
Essentially, the directional limits of the endpoints must match with the function values there. We use the right limit for the left endpoint and the left limit for the right endpoint, which is intuitive.

We need to understand continuity on a closed interval to be able to apply the *Intermediate Value Theorem*.

The Intermediate Value Theorem

The Intermediate Value Theorem allows us to know for certain that there will be a value between two points on a continuous curve.

Let f be a real-valued function continuous on $[a,\ b]$ and differentiable on $(a,\ b)$. Suppose $f(a) \leq K \leq f(b)$. Then there must be a point c on the interval $[a,\ b]$ such that $f(c) = K$



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This may seem like common sense, but it is important to be able to rigorously define such theorems — that is, after all, the foundation of mathematics (and also because the College Board requires you to know the Intermediate Value Theorem, since it is on the curriculum for AP Calculus ® and appears frequently on FRQ's).

Let's see an example:
Prove that the polynomial function $f(x) = x^3 + 2x - 1$ *has a zero in the interval* $[0, 1]$.

We know that all polynomials are continuous on $(-\infty, \infty)$. $f(0) = -1$ and $f(1) = 2$; therefore, by the Intermediate Value Theorem, there must exist a value $x = c$ such that $f(c) = 0$.

Here is how your explanation should be worded in order to earn full points on the AP exam. The <u>9 things</u> graders look for are underlined. <u>Pay close attention to where brackets are open or closed.</u>

"Since f (x) is continuous on <u>$0 \le x \le 1$</u>, <u>$f(0) = -1$</u>, <u>$f(1) = 2$</u>, *and* <u>$-1 < 0 < 2$</u>, *by the Intermediate Value Theorem, there must exist a* <u>*value x = c*</u> *such that* <u>$f(c) = 0$</u>*."*

Remember, if this were an FRQ question, it is crucial that you include all 9 components outlined above.

The Squeeze Theorem

Suppose that we have three fitness-conscious students A, B, and C. A <u>always</u> eats less than or equal to the number calories per day that B eats, and B <u>always</u> eats less than or equal to the number of calories that C eats. In other words, we can express the number of calories consumed by each student in relation to the others using the inequality $A \leq B \leq C$.

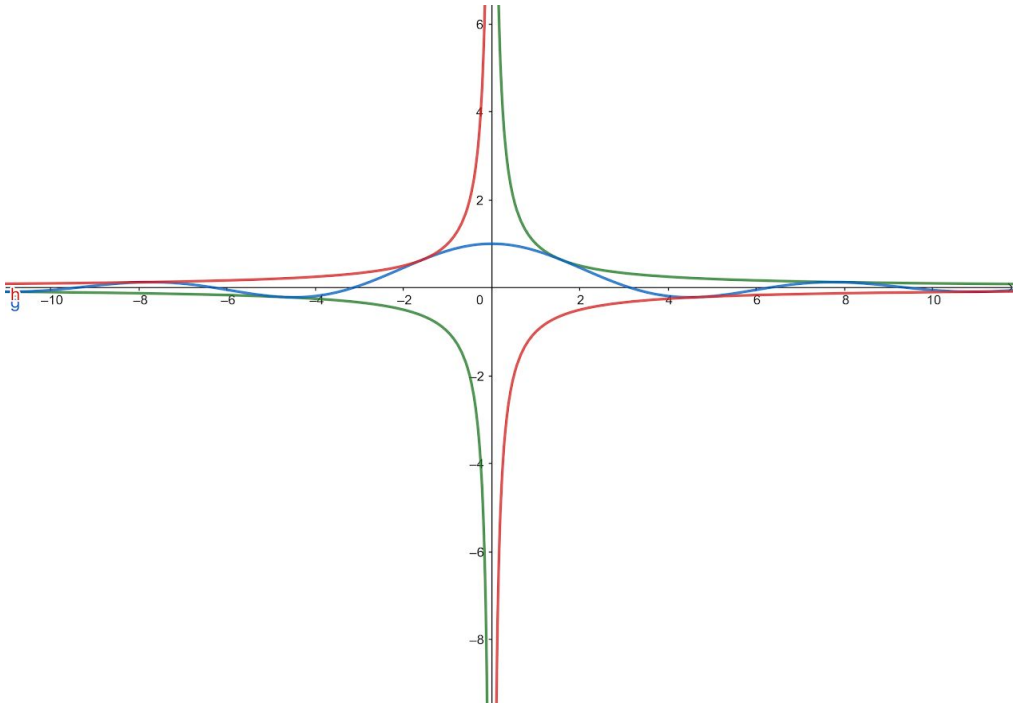
Say that one day, both A and C eat 1600 calories each. Do we know how many calories B consumed on that same day?

Since A must <u>always</u> be <u>less than or equal to</u> B, and since C must <u>always</u> be <u>greater than or equal to</u> B, if A and C have the same value, it follows that B is also equal to that value. So, since we know that $A \leq B \leq C$, if $1600 \leq B \leq 1600$, we know that B must also equal 1600. Therefore, if A and C both eat 1600 calories/day, B must also be eating 1600 calories/day.

The Squeeze Theorem tells us that for real-valued functions f, g and h, if $f(x) \leq g(x) \leq h(x)$ for all x in the neighborhood of c and $\lim_{x \to c} f(x) = \lim_{x \to c} h(x) = L$, it follows that $\lim_{x \to c} g(x) = L$.

While simple, this theorem comes in handy when we are asked to solve limits that we can't solve using conventional algebraic methods.

For instance, how can we solve $\lim_{x \to \infty} \frac{sin\,(x)}{x}$?



Rather than try to solve this limit algebraically, we can try to use the Squeeze Theorem to determine the answer.

First notice that $-1 \leq sin\,(x) \leq 1$. Divide the inequality by x, and we have that $-\frac{1}{x} \leq \frac{sin\,(x)}{x} \leq \frac{1}{x}$.

We know that $\lim_{x \to \infty} \frac{1}{x} = 0$. We also know that $\lim_{x \to \infty} -\frac{1}{x} = 0$.

Therefore, by the Squeeze Theorem, $\lim_{x \to \infty} \frac{sin\,(x)}{x} = 0$.

We can verify the limit by graphing the function, as shown above. Notice that for all real numbers, $-\frac{1}{x} \leq \frac{sin\,(x)}{x} \leq \frac{1}{x}$. Essentially, $\frac{sin\,(x)}{x}$ is bounded above by $\frac{1}{x}$ and $-\frac{1}{x}$.

Exercises:

Compute the following limits. If the limit does not exist, just put *DNE*. You can check your answers at the end of this unit.

Part A.

1. $\lim_{x\to 2} 2x + 5$

2. $\lim_{x\to -1} 5x^3 + 3x^2 + 7x + 1$

3. $\lim_{x\to e} ln\,(x^2) + 1$

4. $\lim_{x\to 1/2} sin^{-1}\,(x)$

5. $\lim_{x\to 3} \frac{3x^2+5x+7}{x}$

6. $\lim_{x\to \pi/2} cos\,(x)$

Part B.

7. $\lim_{x\to 2} \frac{1}{x^2-4}$

8. $\lim_{x\to -4} \frac{x^2+7x+12}{x^3+6x^2+8x}$

9. $\lim_{x\to 4} \frac{x-4}{\sqrt{x}-2}$

Part C.

10. $\lim_{x\to\infty} -7x^2 - 3x^3 + 9x + 1$

11. $\lim_{x\to -\infty} x^4 - 81$

12. $\lim_{x\to\infty} 0.001x^{999999} + 87x^3 - 30x^2 + 7x -$

13. $-\frac{7}{9}(\lim_{x\to\infty} -x^5 - 1)$

14. $\lim_{x\to -\infty} 3x$

15. $\lim_{x\to +\infty} \frac{3x^3+7x+1}{2x^2-7}$

16. $\lim_{x\to -\infty} \frac{\sqrt{x^2}}{x+3}$

17. $\lim_{x\to +\infty} \frac{\frac{2x+7}{3x^3}}{\frac{x^3+7x^2+9}{x}}$

18. Given $\lim_{x\to 2} f\,(x) = 7$, $\lim_{x\to 2} g\,(x) = -3$ and $\lim_{x\to 2} h\,(x) = 9$, compute the following:

    a) $\lim_{x\to 2}[3f\,(x) + \lim_{x\to 2} g\,(x)\,h(x)]$

    b) $\lim_{x\to 2} [(\frac{f\,(x)}{g\,(x)})^2 - log_{\,3}(h\,(x))]$

    c) $\lim_{x\to 2}[\frac{f\,(x)}{h\,(x) - g\,(x)}]$

19. Given $\lim_{x\to 5} f\,(x) = -2$, $\lim_{x\to 5} g\,(x) = \frac{1}{2}$ and $\lim_{x\to 5} h\,(x) = 1$, compute the following:

    a) $\lim_{x\to 5}[f\,(x) + \frac{1}{h\,(x)}]$

    b) $\lim_{x\to 5}[f\,(x) + \frac{h\,(x)}{g\,(x)}]$

    c) $\lim_{x\to 5}[\sqrt{h\,(x)} - 3g(x)\,h(x)]$

20. Given $\lim_{x\to -1} f\,(x) = 27$, $\lim_{x\to -1} g\,(x) = \frac{7}{9}$ and $\lim_{x\to -1} h\,(x) = 0$, compute the following:

    a) $\lim_{x\to -1}[f\,(x) + \sqrt[3]{\frac{7g\,(x)}{9\,h\,(x)+1}}]^{h(x)}$

21. Given $\lim_{x\to 0} f\,(x) = 5$, $\lim_{x\to 0} h\,(x) = -7$, $f\,(-7) = 3$ and $h\,(4) = 0$, solve the following limits.

    a) $\lim_{x\to 0} f\,(h(x))$

    b) $\lim_{x\to 0} h\,(f(x) - 1)$

Answer Key:

Exercise 1.1:

1. 7
2. -8
3. 3
4. $\frac{\pi}{6}$
5. $\frac{49}{3}$
6. 0
7. DNE
8. $-\frac{1}{8}$
9. 4
10. $-\infty$
11. ∞
12. ∞
13. ∞
14. $-\infty$

15. Notice that the highest power in the numerator exceeds that of the denominator. Therefore, $\lim\limits_{x\to+\infty} \frac{3x^3+7x+1}{2x^2-7} = \infty$

16. Divide each term by the highest power: $\lim\limits_{x\to-\infty} \frac{\sqrt{x^2+1}}{x+3} = \lim\limits_{x\to-\infty} \frac{\sqrt{\frac{x^2+1}{x^2}}}{1+\frac{3}{x}} = -1$

17. $\lim\limits_{x\to+\infty} \frac{\frac{2x+7}{3x^3}}{\frac{x^3+7x^2+9}{x}} = \lim\limits_{x\to+\infty} \frac{2x+7}{3x^3} \cdot \frac{x}{x^3+7x^2+9} = \lim\limits_{x\to+\infty} \frac{2x^2+7x}{3x^6+21x^5+27x^3} = 0$

18. a) -6

    b) $\frac{31}{9}$

    c) $\frac{7}{12}$

19. a) -1

    b) 7

    c) $-\frac{1}{2}$

20. a) 1

21. a) $\lim\limits_{x\to0} f(h(x)) = f(\lim\limits_{x\to0} h(x)) = f(-7) = 3$

    b) $\lim\limits_{x\to0} h(f(x) - 1) = h(\lim\limits_{x\to0} [f(x) - 1]) = h(4) = 0$